MILITARY SPECIFICATION

TAMPING MACHINES, RAILWAY, VIBRATOR TYPE, DIESEL-ENGINE-DRIVEN

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of commercial railway tamping machine for ballast tamping, normal track lifting, lining and leveling in railway surfacing including switches, and retimbering operations. The tamping machine shall be of the latest model of the standard product of the contractor and shall have been marketed for 1 year prior to the dates of the opening of the technical proposal. The tamping machine shall be complete with all components that are customarily offered with tamping machines of this type, furnished as standard equipment whether stipulated herein or not, together with such other equipment as may be necessary to enable the tamping machine to function reliably and efficiently in sustained operation, or to meet requirements set forth in this specification.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-A-46153 - Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.

FSC 2230

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of the specification required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for approval shall apply.

AMERICAN RAILWAY ENGINEERING ASSOCIATION (AREA)

Manual of Recommended Practice.

(Application for copies should be addressed to the American Railway Engineering Association, 59 East Van Buren Street, Chicago, IL 60605.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

DEPARTMENT OF LABOR (DOL)

OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

Technical society and technical association specifications and standards are generally available from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 General. The railway tamping machine (hereinafter called "tamper") shall be a heavy-duty, 16-tool, diesel-engine-driven, self-propelled, wheel mounted for 56-1/2 inch gage railroad track and complete with equipment necessary for switch, full face, and spot tamping of ballast.

3.1.1 Performance. The tamper shall perform switch, full face, and spot tamping functions. The tamper shall be capable of track lining, surfacing, and cross-level tamping of main track area and the entire switch.

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The four (min.) vibrators shall be electrically or hydraulically powered and self-lubricating. The tamper system shall have vibratory squeeze-in action. A center turntable for changing direction and set-off shall be provided. The tamper shall be capable of lining, surfacing, and tamping a minimum of 1000 feet per hour of tangent track with no switches. The tamper shall comply with the tamping recommendations for tamping railroad ballast in the AREA manual of recommended practice.

3.2 First article (first-produced tamper). The contractor shall furnish a tamper for evaluation and demonstration within the time frame specified, to prove that his production methods will produce tampers that comply with the requirements of this specification. Evaluation and demonstration shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.2).

3.3 Material. All material shall be as specified herein. Material not specified shall conform to standard commercial practice for this type of equipment (see 6.5).

3.4 Safety. All exposed parts subject to high operating temperatures or are energized electrically shall be insulated, fully inclosed, or guarded. The tamper shall comply with OSHA regulations.

3.5 Engine. The contractor shall furnish a liquid-cooled diesel engine with all accessories offered on his commercial tampers. The engine shall conform to all Federal laws and regulations concerning air pollution control in effect for this type of equipment at the time of contract award. The power and speed rating of the engine shall be such that operation of the tamper under any of the operating conditions specified herein will not require horsepower in excess of the continuous horsepower rating of the engine as previously published by the engine manufacturer, or 85 percent of the engine manufacturer's published maximum horsepower rating, at the applied speed, whichever is less.

3.6 Mounting. The tamper shall be mounted on four single flanged wheels up to 24 inches in diameter. The drive axle shall be a heavy-duty type. Wheel tracking width shall be suitable for use on 56-1/2 inch gage railroad track. The frame shall be constructed of structural steel and shall include traversing rails for support and positioning of the tamper workheads.

3.6.1 Suspension. The tamper shall be shock dampened between the axles and the frame.

3.7 Propelling unit. The tamper shall be self-propelled by means of a hydraulic motor driven by one of the hydraulic pumps. The motor shall drive a transmission that provides three (min.) forward speeds and three

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(min.) reverse speeds to the drive axle. The tamper shall be capable of traveling 20 mph in either direction while in the highest gear.

3.7.1 Multi-wheel drive. The tamper shall be equipped with a two-wheel front drive when 70 percent or more of the tamper's weight is located on the two front wheels. However, the contractor shall furnish a four-wheel drive when less than 70 percent of the weight is located on the two front wheels.

3.8 Brakes. Service brakes shall be air operated, friction shoe type for all wheels. Provisions for parking and emergency stopping shall be furnished. Parking brake shall securely hold the tamper from rolling on grades normally encountered in railroad maintenance.

3.9 Hydraulic system.

3.9.1 Hydraulic fluid reservoir. A hydraulic fluid reservoir shall be provided and shall have sufficient capacity to hold the required volume of hydraulic fluid, with not less than 10 percent for expansion space. The reservoir shall be provided with a filler opening, a removable magnetic oil strainer, and a filler cap, accessibly located to facilitate filling the tank. Provisions shall be made for draining of the reservoir and determining the level of the fluid. The reservoir drain shall be so located that the fluid will not drain on other components of the unit.

3.9.2 Hydraulic oil filter. Manufacturer's standard hydraulic oil filtration system shall be furnished.

3.9.3 Hydraulic pump(s) system. A hydraulic pump(s) with necessary pressure and flow capacity to operate all hydraulic powered components including track lining and vibratory squeeze-in devices, shall be provided. The hydraulic pump(s) shall be driven by heavy-duty power takeoff units. Chain drive or belt shall not be used. The pumping system shall include an adjustable pressure relief valve or unloading valve with the necessary range of adjustment.

3.9.4 Hydraulic cylinders. All hydraulic cylinders shall incorporate seals to prevent entrance of moisture and dirt into the cylinders and leaks of hydraulic fluid from the cylinders.

3.9.5 Hydraulic lines and fittings. Hydraulic lines shall be located to facilitate maximum ease of removal and installation, or visual inspection for failure and maximum protection from damage during operation of the boom assembly. Flexible hydraulic hose or collector rings shall be used where flexing of hydraulic lines is encountered, and where installation of components requires a flexible connection. Hydraulic lines and fittings shall be installed to preclude vibration and chafing. Hose assemblies and all

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seamless steel hydraulic lines shall be of adequate size to provide the rate of flow, temperature, and pressures required to operate the system smoothly and efficiently.

3.10 Turntable. A built-in hydraulically actuated center mounted turntable shall be provided for changing travel direction and for set-off purposes. Turntable shall be rail contacting type. The propelling system shall be canceled automatically while the turntable is operating.

3.11 Tamping system. Tamping system shall consist of a traversing carriage assembly of two hydraulically positioned workheads. Each workhead shall consist of two (min.) vibrators. Each vibrator shall be equipped with two (min.) cast or forged steel tamping tools. The total system shall consist of 16 tools. Vibrator frequency shall be 3,200 (min.) vibrations per minute. Vibrator workhead shall be hydraulically positioned to vary tamping tool working depth to 19-1/2 inches (nominal) below the top of the rail. Downward force of each workhead, without tamping tools, shall be 6,300 (min.) pounds. Vibrator workheads shall be operable independently and in unison.

3.11.1 Tamping system controls. Tamping system controls shall be electro/hydraulic type and equipped for fully automatic operation. Automatic operation shall control the following after setup:

(a) Tamping to preselected depth.
(b) Concentration of ballast to a preselected pressure.
(c) Jacking to a preselected height.
(d) Elevation of workheads and jacks to a preselected height before indexing to the next tie.

Placement of the controls shall be so as to prevent inadvertent actuation by the operator.

3.12 Lining, surfacing, and cross-leveling. The track lining and surfacing systems shall have sensing devices to detect and transmit misalinement data (both horizontal and vertical) to the measuring, lining, and surfacing devices. The machine shall be so controlled that the misalinement information is automatically received and the track is automatically lined, surfaced, and cross-leveled (or super-elevated on curves). The machine shall be capable of lining track to the above tolerances in one pass if the track is from 0 to 5 inches out of line on either tangents or curves. The machine shall be capable of surfacing and cross-leveling or super-elevating track (either spot surfacing of raising any required amount from 0 to 5 inches) in one pass if the track is from 0 to 5 inches out of surface and 0 to 5 inches out of cross-level. The machine shall be equipped with a digital or graphical read-out device to indicate curve offsets or versines.

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3.13 Operator's compartment. Operator's compartment shall be fully enclosed, sound dampened, heated, and air conditioned. Adequate safety windows to provide an unobstructed view of operations shall be provided. Compartment shall be equipped with an inside light and shall have forward and rearward facing traveling lights mounted outside the cab.

3.14 Air system. Manufacturer's standard air system shall be furnished complete with all accessories.

3.15 Lubrication. Lubrication shall be in accordance with the manufacturer's recommendations. Pressure lubrication shall not damage seals or other parts. All parts requiring lubrication shall be lubricated prior to delivery and tagged to show the type and temperature rating of the lubricant used. Hydraulic lubrication fittings shall be in accordance with SAE Handbook.

3.16 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants, such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion products, or any other interfering substances. As soon as practicable after cleaning, and before any corrosion product or other interfering material can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to the surface as soon as practicable after cleaning and treating. Painting shall be with the manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall not be less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects. Color of the finish coat shall be yellow as specified in AREA manual of recommended practices.

3.17 Instruction plates. The tamper shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws or bolts.

3.18 Identification plate. Each tamper shall be provided with a plate, made of noncorrosive metal, on which shall be stamped; letters, numerals, and other characters of such size as to be clearly legible, the following information:

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Nomenclature	Shipping Date
Serial Number	Shipping Weight
Model	Shipping Cubage
Contract Number	

3.19 Workmanship. Workmanship for the construction and fabrication of the tamper shall be in accordance with standard commercial practice for this type of equipment and the American Railway Engineering Association (AREA) manual on construction and maintenance of way equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections (evaluation and demonstration) set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First-produced tamper evaluation. The first-produced tamper shall be compared with the requirements of this specification, and the contractor's published literature and calculations to verify conformance to this specification. The tamper shall be subjected to the following demonstration, and failure of any portion of demonstration shall be cause for rejection of the tamper.

4.3 First-produced tamper demonstration. The first-produced tamper, in accordance with this specification, shall be subjected to an operational demonstration which shall consist of tamping ballast around and under 300 ties. The tamper shall be operated over an additional length of track with a switch such that the tamper lining, surfacing, and cross-leveling functions are demonstrated. Inability of the tamper to conform to requirements of this specification, any malfunction, or any leakage of fuel or lubricants shall be cause for refusal of acceptance of the tamper by the Government until the problem is corrected. The contractor shall submit a report to the Government upon conclusion of the demonstration (see 6.2). This report shall include a description and results of the demonstration performed, including any malfunctions, and actions taken to correct such malfunctions.

4.4 Run-in. Each tamper (except first-produced) shall be subjected to the contractor's standard run-in procedure, except that the run-in shall be for not less than 2 hours of operation. Any malfunction or any leakage of fuel or lubricants shall be cause for refusal of acceptance of the tamper by the Government until the problem is corrected.

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5. PACKAGING

5.1 Preservation-packaging. Unless otherwise specified (see 6.2), the tamper and all components and attachments shall be preserved and packaged for domestic shipment in accordance with the contractor's normal commercial practice. Items with critical surfaces required for fit or function shall be provided protection by the use of preservative coatings. Items requiring protection from physical damage shall be protected by wrapping or cartonizing of the individual item. All lubrication reservoirs, sumps, etc., shall be filled to the operating level with the grade of military lubricant designated for use in the temperature range at the shipping destination. The cooling system shall be filled with a fresh clean solution of 50 percent water and 50 percent antifreeze conforming to MIL-A-46153. Two tags shall be attached in a conspicuous location: One to indicate which lubricant has been used, the temperature range of lubricant used and grade of lubricant used, and the other to indicate the temperature to which the cooling system can be subjected to before freezing. A deprocessing guide shall be furnished indicating in detail the preservation materials to be removed and any other servicing required prior to placing the tamper in operation.

5.2 Marking. Marking for commercial packaging shall reflect the item name, National Stock Number, quantity, and contract number.

6. NOTES

6.1 Intended use. The tamper is intended for packing ballast around and under railway ties, and lining, surfacing, and cross-leveling track.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame for submitting first-produced tamper (see 3.2).
(c) Time frame for submitting first-produced demonstration report (see 4.3).
(d) When the tamper shall be preserved and packaged for other than domestic shipment (see 5.1).

6.3 First-produced tamper. Any changes or deviations of production tampers from the approved first-produced tamper during production will be subject to the approval of the contracting officer. Approval of a first-produced tamper will not relieve the contractor of his obligation to furnish tampers conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each tamper.

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6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-T-52954(ME) Tamping Machines, Railway, Vibrator Type, Diesel-Engine-Driven

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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